

DSC 102: Systems for Scalable Analytics

Programming Assignment 5: Grading Scheme

Programming Correctness (100)

For each task, we will run several tests on it with our hidden datasets. Your code must pass all the tests to be counted pass for the task.

Task No.	Task Description	Score (pass/fail)
1	Combine tables and group-by aggregations	15/0
2	Flatten schema and handle array and map type	10/0
3	Flatten schema and conduct self-joins	20/0
4	Typecasting and data imputation	10/0
5	Apply word2vec on string data	10/0
6	One-hot encoding and PCA on categorical data	15/0
7	Train a decision tree regression model	5/0
8	Hyperparameter tuning for the decision tree regression model	15/0

Timeout: We will run your code (all eight tasks put together running one after another) on a three-worker cluster and impose a timeout of 2 hours and will kill your process at that point. Only tasks that finish within the timeout will be considered for the correctness portion. If any of the tasks fails, you will still get partial credits for the other tasks. We will also deduct points if it takes longer as per the following.

- Between 60 min and 90 min: -10 points
- Over 90 min: -30 points

For instance, if your code took 75 min to run and it only passed Tasks 1-4, your final score will be $45 = 15$ (Task 1) + 10 (Task 2) + 20 (Task 3) + 10 (Task 4) - 10 (Overtime penalty).

Extra Credit (10)

Your code will be timed with all tasks together. If it manages to pass all the tests for Programming Correctness, you may receive extra credits as showed in the following table according to the runtime.

Runtime t	Credits
$t \leq 15$ min	10
$15 \text{ min} < t \leq 40$ min	5